TMF Health Quality Institute is a nonprofit organization that works with government agencies, providers and other groups to improve health care quality and outcomes. TMF is headquartered at 3107 Oak Creek Drive #200 in Austin, TX. Contact: tara.richardson@tmf.org

PC-1. What health management or care navigation apps would help you understand and manage your (or your loved ones) health needs, as well as the actions you should take?

Patients and caregivers need to be able to access health information in electronic form using applications that are complimentary and easy to use. These applications should be accessible using computers and mobile devices and should be available with any operating system. Patients and caregivers should be able to use these applications to access their health information including, but not limited to, care summaries, medication lists, hospital stay reports, and diagnostic reports. Instructions and educational resources related to their health information should also be included. Patients and caregivers should have applications available to them that integrate with the various EHRs, such as Epic's MyChart, the Oracle Health Patient Portal, and Meditech's MHealth app.

a. **What are the top things you would like to be able to do for your or your loved ones' health that can be enabled by digital health products?**

   Many patients and caregivers have a difficult time learning about their test results. They need assistance accessing these results, whether it be through an application or even getting the information directly from their provider(s). While they wait on their results, they are anxious and stressed. Communications between patients and providers may be improved by using an optimized EHR as well as AI. Patient portals are a great way to improve communications; however, more work should be done. Many times, patients are sent a link to register for a patient portal without much instruction. Providers should provide complete instructions and assistance to patients and caregivers and follow up with them to ensure they have been enrolled and can access their information successfully. This may mean that staff needs to sit with the patient and walk through the various workflows in order to ensure they are able to use the application to the fullest extent. Patients and caregivers should be able to use applications to schedule/cancel appointments, make payments, ask questions about their billing statement, and ask provider questions they have about their care summaries, medications, and diagnoses. They should be able to not only receive communications from staff, but their primary care provider and specialists as well.

b. **If you had a personal assistant to support your health needs, what are the top things you would ask them to help with? In your response, please consider tasks that could be supported or facilitated by software solutions in the future.**

   Currently, there are many digital applications available for mobile devices. Patients and caregivers should be able to use an application of their choosing with the ability to send and receive data seamlessly with a provider's EHR. Families want to use technology to be proactive in their health management in order to prevent chronic conditions.

Applications, such as MyFitnessPal, which can be used to track fitness goals, caloric intake, nutritional values, exercise, and sample meal plans have been found to be helpful for users. The value would increase even more if patients and caregivers could use applications such as this to send the data to providers for them to review and monitor.

Other applications found to be helpful in managing conditions and reduce further complications include Medisafe, which allows users and caregivers to monitor medication adherence, sends notifications of potential adverse reactions, and supports trackers such as glucose and cholesterol. Refill and appointment reminders help patients and caregivers manage adherence. Headspace and Calm are apps that help users with their mental health and wellness by providing guided methods for stress and sleep improvement as well as mental well-being.

PC-2. Do you have easy access to your own and all your loved ones' health information in one location (for example, in a single patient portal or another software system)?

a. If so, what are some examples of benefits it has provided?

Some patients and caregivers are able to access their health information in one patient portal. Patient portals that integrate with the EHR, such as Athenahealth, for example, have allowed the ability to view billing information and pay bills. They also provide the ability to schedule and manage appointments online verses having to call the provider's office. Users are able to view care summaries, health history, and current medications. One of the main benefits is being able to communicate with the provider and staff more easily and quickly since electronic messages can be sent back and forth.

b. If not, in what contexts or for what workflows would it be most valuable to use one portal or system to access all such health information?

Patients and caregivers may have a difficult time tracking their health information, especially if they see multiple providers. Imaging and lab tests may be ordered by one provider, but those results are not sent to all providers. So, if a patient is being seen by their Primary Care Provider as well as a specialist and each system is not set up to send or receive the information, then information may be missed. Patients who see change providers may experience problems when trying to sign up for a patient portal if the new provider uses the same EHR. The EHR expects a new email to be submitted so it rejects the registration. Providers and staff have a difficult time resolving this issue for patients; therefore, the patient may not register for the new provider's portal. EHR vendors should work to resolve this issue to reduce barriers.

Some providers may have a patient portal; however, patients and caregivers don't always have the ability to sign up for the portal on their own for various reasons, and they may need assistance. Also, they may not have the trust needed to sign up and complete the setup process. There may be a trust issue with the provider or the application. Patients don't always trust having their information on the internet or in the cloud, so they may not be willing to use the portal. Therefore, they may continue to struggle accessing their

health information. Ideally, patients and caregivers would be able to access all of their health information in one, standardized, application. This would decrease confusion and improve engagement in their healthcare.

c. **Were there particular data types, such as x-rays or specific test results, that were unavailable? What are the obstacles to accessing your own or your loved ones' complete health information electronically and using it for managing health conditions or finding the best care (for example, limitations in functionality, user friendliness, or access to basic technology infrastructure)?**

   Patients experience barriers to accessing x-rays and other test results on a regular basis. The data isn't easily accessible and so they wait and become anxious about possible diagnoses and conditions. This increases the burden on providers as staff then need to manage phone calls between the patient or caregiver and the provider. The inability to access test results causes repeated tests, which is not cost effective and increases burden for patients. Patients and caregivers should be able to look up test results in the patient portal more easily. Systems should be integrated with the ability to send and receive messages in various data formats, which would improve workflow for providers. This would lead to increased interoperability, allowing patients to view their information in order to plan. Many patients are referred to specialists but don't really know how to proceed afterwards. They should be able to log in, view their results, and decide whether or not they need to call their provider with questions or wait until their follow up visit to review with the provider. If providers were able to participate in a Health Information Exchange (HIE), data would be more accessible and information flow would be more seamless. Providers and therefore, patients, would no longer have to hunt and dig for the information they need to manage their conditions and/or prevent long-term health issues.

PC-3. Are you aware of health management, care navigation, or personal health record apps that would be useful to Medicare beneficiaries and their caregivers?

There are many applications available for patients and caregivers to use to manage their health information. Some include MyID, Apple Health, My Health Records, MyChart, and CommonHealth.

Patient intake systems, such as Phreesia, can improve information exchange and workflows for providers and improve communication with patients when integrated with EHRs, such as Athenahealth. Phreesia allows patients the opportunity to complete electronic forms and provide financial and insurance information as well as clinical information including health history, current conditions, and medication details. Once they complete the check-in process, the information flows into the provider's EHR. Staff are able to use the information to set up the patient's chart, which can then be viewed by clinical staff in order to complete the assessment during the patient's visit with the provider. Intake systems like Phreesia can be used in any provider setting making workflows more seamless for patients and providers. These systems can be used for visit reminders, post visit patient experience surveys, and provide educational information about conditions and potential treatments, which helps patients stay engaged in their care.

PC-4. What features are missing from apps you use or that you are aware of today?

Applications are missing health information integration among the various EHRs and third-party applications. Patients who have multiple providers may have to use more than one application. Some of the applications do not include the patient's full comprehensive health history. They don't pull in all pieces of data, so the patient is left to manually enter the data themselves into the application. This discourages patients from using the application as well as trusting the system. Ease of use is another problem with some applications. Patients who find the app hard to use or "clunky" and not intuitive will not be willing to use the app. Another issue is that apps typically don't allow the patient to edit any of the information. The patient may have to call the provider in order to get the information updated, which takes time and is not efficient. Applications should allow the patient to edit their own health information, which should then be reviewed by the clinical staff/provider. Another issue patients have is that they don't see a complete medication history. They may find that not all of their medications are listed, or they may find that a medication that they stopped taking years ago is still in the system. Providers experience problems as well when trying to reconcile medications. Even though they may reconcile medications and remove those that have been discontinued, they appear in the system at the patient's next visit. This causes mistrust in the system and frustration for both patient and provider. Programming improvements should be made in data validation.

a. **What apps should exist but do not yet? Why do you believe they do not exist yet?**

   We should have an app that includes information from our entire medical record, including diagnostic test results, physical and occupational therapy notes and treatment plans, assessments and progress notes from primary care and specialists. Patients and caregivers should be given a personal card and/or have this information available through technology built into a driver's license. Providers of all types should be able to scan this card and view the information for registration, treatment, and billing (e.g., payments). Patients and caregivers should be able to use digital applications on their mobile devices and then download this information to the card and/or have the information integrated from the application to the EHR so that everything involving the patient's health is seamless and interoperable.

b. **What set of workflows do you believe CMS is uniquely positioned to offer?**

   It would be best if the patient could access all health information from their primary care physician, specialists, and hospitals in one easily accessible, user friendly, and complimentary application.

   CMS could potentially allow patients to access their claims data, medication detail from pharmacies, and other health information, making this information accessible to all beneficiaries and appropriate providers. Doing so would allow patients, caregivers, and providers the ability to view information in one location, which would increase interoperability, enhance decision making, and improve patient safety.

   Integrating payer data such as demographics, quality measures, and claims could increase patient safety, improve patient care, and close care gaps.

PC-5. What can CMS and its partners do to encourage patient and caregiver interest in these digital health products?

CMS could encourage the use of digital health products by collaborating with its Quality Innovation Network-Quality Improvement Organization (QIN-QIO) contractors to expand the knowledge patients and caregivers have about their overall health. Patients and their caregivers usually trust their healthcare team more than governing bodies, so if they see that their providers are using technology to enhance their own processes, they will be more willing to use it themselves. Deploying QIN-QIOs and providers to training sessions held in the community where patients feel more comfortable, would build trust and help build relationships with CMS, providers, and patients.

Along with the importance of having digital products that are easy to use available for their health management, patients need to understand that their health information will be protected. CMS and QIN-QIOs could provide easy to understand information on data security and compliance standards to help build trust in the technology and increase adoption.

Providing complimentary collateral (advertisements and educational materials) for providers to distribute and display in their care settings can be helpful. These items should also be available for download by providers, patients, and caregivers from the CMS website.

Any improvements made to Blue Button should be well advertised so that all beneficiaries are aware. Tangible benefits, such as time savings, increased communication, and improved outcomes should be celebrated and advertised to increase awareness about the benefits of digital health technology.

a. **What role, if any, should CMS have in reviewing or approving digital health products on the basis of their efficacy, quality or impact or both on health outcomes (not approving in the sense of a coverage determination)? What criteria should be used if there is a review process? What technology solutions, policy changes, or program design changes can increase patient and caregiver adoption of digital health products (for example, enhancements to data access, reimbursement adjustments, or new beneficiary communications)?**

   Digital health products could be certified in the same manner as EHRs are certified under the ONC Health IT Certification Program. This would help ensure apps meet current regulations such as HIPAA, FDA regulations, 21st Century Cures Act & ONC Information Blocking, and the FTC Health Breach Notification Rule. Any other third parties working with the vendors of the digital health products should be compliant with these regulations as well.

b. **What changes would enable timely access to high quality CMS and provider generated data on patients?**

CMS could collaborate with QIN-QIOs and other organizations across the United States to ensure that broadband is available in all areas, including small rural areas. Broadband initiatives, such as the Broadband Equity, Access, and Deployment (BEAD) Program, Affordable Connectivity Program (ACP) and the Tribal Broadband Connectivity Program are important programs that are necessary funding opportunities for many providers. CMS could continue to work with the private sector as well as internet service providers to help make broadband affordable in order to increase data exchange and improve interoperability.

PC-6. What features are most important to make digital health products accessible and easy to use for Medicare beneficiaries and caregivers, particularly those with limited prior experience using digital tools and services?

Digital health products are easier to use when they are intuitive and have a visually pleasing user interface. The user interface should have text that is larger or that can be adjusted so that users are able to read the text on the screen easily. They should also be easy to set up and offer guided instructions and chat features so that users can easily ask for help "on the fly". Notifications and messages should be easy to manage so that users do not become overwhelmed. There should also be shared access options so that caregivers can manage their loved one's health information more easily. Users should also have the option to install applications on multiple devices in order to access their health information in different locations and in different ways. All of these applications should have security measures in place so that the information is well protected. One important feature that should be included is the ability for the user to request a password reset. This should be automated functionality so that the user can make the request electronically instead of having to call the provider office and ask for manual assistance. This would greatly ease user and staff frustration and improve efficiency. Consents should be easy to read and supported by secured infrastructure.

PC-7. If CMS were to collect real-world data on digital health products' impact on health outcomes and related costs once they are released into the market, what would be the best means of doing so?

CMS could use assistance from its QIN-QIO partners to seek feedback and participation from patients and caregivers to assess the impact of digital health products on health outcomes. Doing so can shape solutions that reflect real-world needs.

Multiple data collection and analysis methods could be employed. This may include a structured, digitized survey that captures responses electronically so that analysis can be performed. Insurance claims and EHR data can be utilized to track patient outcomes and associated costs. Sources of information may include data collected from remote monitoring and wearable devices where information is exchanged between the devices and the EHR.

Clinical and financial data and claims data could be used to conduct cost-benefit analyses and quality-adjusted life years (QALY) assessments to measure the overall impact of digital health products. Leveraging advanced analytics and AI-driven insights would enhance the effectiveness of data interpretation and decision-making.

PC-13. How can CMS encourage patients and caregivers to submit information blocking complaints to ASTP/ONC's Information Blocking Portal? What would be the impact? Would increasing reporting of complaints advance or negatively impact data exchange?

CMS can take a number of steps to raise awareness of information blocking:

- Require providers to include a description of information blocking and how to report it when they distribute materials to patients
- Require signage in hospitals, clinics, nursing homes, and other healthcare delivery environments
- Run an awareness campaign, distributing information through a variety of channels (e.g., TV, news outlets, social media, email, online)
- Include a brief informational section about information blocking on patient-facing CMS web pages

Increased awareness and reporting will provide additional incentives for providers to be effective in sharing information.

PC-14. Regarding digital identity credentials (for example, CLEAR, Login.gov, ID.me, other NIST 800-63-3 IAL2/AAL2 credentialing service providers (CSP)):

a. **What are the challenges today in getting patients/caregivers to sign up and use digital identity credentials?**

   There are technological barriers. Not everyone has a smartphone, tablet, or device that is capable of working with the app or website to validate identity. In some cases, only a smartphone can be used, for example. Older patients will no doubt struggle with the process, and even for a technically savvy person, it can take a while to get an identity established.

b. **What could be the benefits to patients/caregivers if digital identity credentials were more widely used?**

   The primary benefit would be secure and ready access to information with one set of credentials.

c. **What are the potential downsides?**

   Many people may not register.

d. **How would encouraging the use of CSPs improve access to health information?**

   It will provide secure access to information and in the long run it will be easier for people. The barrier to entry is a little high, but once over that, people will be able to obtain information more easily with that one set of credentials.

e. **What role should CMS/payers, providers, and app developers have in driving adoption?**

CMS and its partners should continue to provide solutions that are easy to use, provide education, and provide assistance for those who need it.

f. How can CMS encourage patients to get digital identity credentials?

CMS could provide videos on how to go through the process and provide assistance.

PR-1. What can CMS and its partners do to encourage providers, including those in rural areas, to leverage approved (see description in PC-5) digital health products for their patients?

a. What are the current obstacles?

Rural healthcare providers may lack access to reliable broadband internet and up-to-date technology, making it difficult to support telehealth, remote monitoring, and other digital health tools. These limitations may affect both provider operations and patients' ability to participate in virtual care from home.

Inconsistent or unclear reimbursement policies for digital health services discourage adoption. Fee-for-service models often don't account for the time and effort required to implement digital tools.

Providers, especially in rural settings, may not be trained in how to evaluate or implement digital health products. Without clear clinical guidelines, providers may be hesitant to recommend or rely on digital tools in patient care.

Smaller healthcare organizations often operate with limited financial and human resources. They may lack IT support, administrative capacity, or staffing flexibility to invest in, adopt, and maintain digital solutions.

Rural patients may lack access to smart devices, internet connectivity, and the digital literacy needed to engage with digital health tools. Cultural mistrust of technology and concerns about privacy and data security can hinder patient uptake and sustained use.

The Medicare patient population, in both rural and urban areas, tends to hold mistrust in digital applications, making it harder for providers to implement and sustain use. An option for providers could be to partner with community health workers and/or volunteers (e.g. college students) who have been trained on the digital health products who can then sit with and train patients and caregivers on how to use them. This would decrease burden on providers and staff by building patient confidence when using the technology.

b. What information should providers share with patients when using digital products in the provision of their care?

When using digital health products, providers should explain their purpose, how they work, any limitations, and how patient data will be protected.

c. What responsibilities do providers have when recommending use of a digital product by a patient?

- Clinical appropriateness
- Monitoring and acting on the clinical data

PR-2. What are obstacles that prevent development, deployment, or effective utilization of the most useful and innovative applications for physician workflows, such as quality measurement reporting, clinical documentation, and billing tasks? How could these obstacles be mitigated?

Fragmented health IT systems and poor interoperability between EHR platforms make it difficult to integrate new tools seamlessly. Complex and inconsistent documentation requirements increase administrative burden and reduce time for patient care. Smaller or rural practices may lack the financial means, technical support, or incentives to implement solutions effectively. Streamlining and aligning documentation requirements across payers would reduce redundancy. Involving frontline clinicians in the design and testing of workflow tools can improve usability and relevance. Use the QIN-QIOs to expand support for small and rural providers.

PR-3. How important is it for healthcare delivery and interoperability in urban and rural areas that all data in an EHR system be accessible for exchange, regardless of storage format (for example, scanned documents, faxed records, lab results, free text notes, structured data fields)?

The ability to exchange EHR data in various formats in urban and rural areas is imperative in order to deliver high quality care.

Please address all of the following:

a. **Current challenges in accessing different data formats.**

   Having data in various formats such as scanned documents and images, structured fields, and free text notes, creates challenges in data integration and interoperability. There are also issues with proprietary EHR platforms, which increases barriers for providers to exchange data among healthcare teams since formatting cannot be edited. These challenges cause a disruption for users because they have to piece together information from different formats and locations.

b. **Impact on patient care quality.**

   When data is more easily accessible, providers can make better informed decisions. This then reduces medical errors, reduces waste in the system by not repeating labs or x-rays and improves care coordination. The ability to easily exchange data improves communication among the healthcare team, which then improves patient outcomes. Providers can monitor their patients' health and coordinate timely interventions when necessary.

c. **Technical barriers to full data accessibility.**

   There are currently many technical barriers because EHRs have different data structures and standards. Proprietary systems decrease the ability for the different systems to

exchange data and share information. Because of the costs associated with implementing new EHRs, many organizations continue to use legacy systems. These systems are unable to support current interoperability standards. Many organizations use scanned documents for providing and storing information in the patient's record. These documents have limited usability and increase difficulties in data exchange. Currently, providers have restrictions on how to use funds available from CMS. If these restrictions could be removed, providers would have an opportunity to use more funds towards technology upgrades as well as participate in HIEs. The ability to participate in an HIE and exchange data would broaden their capabilities in managing patients' health. Clinicians and patients would be more knowledgeable and more proactive since information could flow back and forth seamlessly.

d. Cost or privacy implications of making all data formats interoperable.

There are high implementation costs associated with upgrading systems to fully support interoperability. Investment in software, infrastructure, and training often deter providers from EHR upgrades. Addressing privacy is another concern providers face since they must ensure their systems are HIPAA compliant. Systems must protect patient privacy while also protecting the organization against cyber-attacks, ransomware, phishing scams, and improper/unauthorized data sharing.

e. Priority level compared to other interoperability needs.

Interoperability is a focus for providers, however, priorities are different between urban and rural organizations. Providers in urban areas may be at the point where they are ready to focus on interoperability and optimization of their systems. Rural providers on the other hand may be struggling with gaining access to broadband and basic connectivity. They may be facing a longer lead time towards interoperability and may just be on the verge of providing electronic access for patients to their health information. All providers must deal with the needs of data security and compliance as well as implementing common standards like HL7 FHIR. They face other priorities including financial and operational necessities. These challenges and competing priorities are likely compounded for small and/or independent providers, who often don't have staff and resources available to support this work once it is available. In order to improve implementation of interoperability in the health ecosystem, providers need support from CMS and QIN-QIOs such as learning systems, particularly for rural, small and independent practices. For example, centralized or external learning support, such as coaches, toolkits or learning events, can support practices needing extra support to find solutions and workflows that work in their environments. Hearing from industry experts as well as peers, through learning systems, can enable all providers, but particularly under-resourced practices, in that final mile of implementation and enable participation in data exchange as well as value-based care.

PR-4. What changes or improvements to standards or policies might be needed for patients' third-party digital products to have access to administrative workflows, such as

auto-populating intake forms, viewing provider information and schedules, and making and modifying an appointment?

In order for this to occur, Fast Healthcare Interoperability Resources (FHIR) standards related to administrative workflows should be improved in order to facilitate real time data exchange. HIPAA regulations need to be more comprehensive. Security enhancements such as multi-factor authentication for third-party apps, data encryption, and audit trails must be required and in place for all systems. There also needs to be improvements made towards gaining consent so that patients know exactly who has access to their data. Providers could receive assistance from QIN-QIOs to ensure these improvements are implemented and optimized.

PR-9. How might CMS encourage providers to accept digital identity credentials (for example, CLEAR, ID.me, Login.gov) from patients and their partners instead of proprietary logins that need to be tracked for each provider relationship?

a. **What would providers need help with to accelerate the transition to a single set of trusted digital identity credentials for the patient to keep track of, instead of one for each provider?**

   They would need to provide simple instructions and assistance to those who need it.

b. **How might CMS balance patient privacy with convenience and access to digital health products and services that may lead to significant improvements in health?**

   Using digital identity credentials will simplify access in the long run, but there is an initial barrier for creating the identity initially.

PR-10. Regarding digital identity credentials (for example, CLEAR, Login.gov, ID.me, other NIST 800-63-3 IAL2/AAL2 CSPs):

a. **What are the challenges and benefits for providers?**

   It will be challenging to assist older patients with initial onboarding. The benefit will be that there is one common set of identity providers for all, so as time goes on, more and more people will already have their credential established and that will simplify the new patient onboarding.

b. **How would requiring their use improve access to health information?**

   There will be a higher degree of assurance that the person who is accessing the patient data is the actual person. Better identity proofing and secure access are the primary benefits.

c. **What are the potential downsides?**

   Difficulty for elderly patients and others who may struggle with technology or not have a device that will work with the process.

e. **How could digital identity implementation improve provider data flow?**

It will make it easier to positively identify the correct patient when sharing information between providers.

PR-14. How can CMS encourage providers to submit information blocking complaints to ASTP/ONC's Information Blocking Portal? What would be the impact? Would it advance or negatively impact data exchange?

CMS and its partners (e.g., QIN-QIOs) could continue to do outreach to providers about the importance of information exchange and encourage them to report. This will incentivize adoption across all healthcare entities.

TD-1. What short term (in the next 2 years) and longer-term steps can CMS take to stimulate developer interest in building digital health products for Medicare beneficiaries and caregivers?

Short-term steps:

- Create developer-friendly data sandboxes with realistic and real-time (as possible) Medicare beneficiary data patterns for testing applications before live deployment
- Establish clear certification pathways that provide developers confidence their products will integrate with existing Medicare systems
- Launch hackathons and innovation challenges, etc. focused on specific Medicare beneficiary needs (chronic disease management, care transitions, medication adherence)
- Provide technical assistance and mentorship programs connecting experienced health IT developers with those new to Medicare populations

Longer-term steps:

- Develop reimbursement models that are outcome based that compensate for digital health interventions showing measurable improvements in quality metrics
- Create interoperability requirements that ensure new digital products can seamlessly integrate with existing Medicare quality reporting systems
- Establish partnerships with QIN-QIOs to provide real-world testing environments and clinical expertise during product development

TD-2. Regarding CMS Data, to stimulate developer interest—

a. **What additional data would be most valuable if made available through CMS APIs?**
    - Real-time quality measure data tied to specific providers and facilities
    - Care gap identification data showing missed preventive services including Annual Wellness Visit gaps, medication adherence issues, and care coordination opportunities
    - Risk stratification data such as HCC risk scores and clinical risk indicators helping identify high-risk beneficiaries who would most benefit from further interventions
    - Provider performance benchmarks to enable comparative effectiveness applications

d. **What other APIs should CMS and ASTP/ONC consider including in program policies to unleash innovation and support patients and providers?**
   - Quality Measure Results API providing standardized access to HEDIS, CMS Star Ratings, and quality reporting data
   - Care Gap Identification API that identifies specific improvement opportunities for individual beneficiaries
   - Provider Network and Performance API combining directory information with quality and cost performance metrics

TD-3. Regarding digital identity implementation:

a. **What are the challenges and benefits?**

   This could be challenging to those who are elderly and/or less tech savvy and/or who do not have devices that will allow for identity proofing.

b. **How would requiring digital identity credentials (for example, CLEAR, Login.gov, ID.me, other NIST 800-63-3 IAL2/AAL2 CSPs) impact cybersecurity and data exchange?**

   It will improve the process by making it more secure and providing a higher degree of assurance that the correct patient is being linked to the data.

c. **What impact would mandatory use of the OpenID Connect identity protocol have?**

   It will make it easier for all to connect to the various patient portals and other services without having to re-establish a username/password.

VB-1. What incentives could encourage APMs such as accountable care organizations (ACOs) or participants in Medicare Shared Savings Program (MSSP) to leverage digital health management and care navigation products more often and more effectively with their patients? What are the current obstacles preventing broader digital product adoption for patients in ACOs?

Incentive	How It Would Operate	Why It Matters for ACO/MSSP Participants
Shared-Savings Accelerator Bonus	Incremental-percentage add-on to earned savings when the ACO meets digital-engagement thresholds (Remote patient monitoring (RPM) or patient reported outcome (PRO) enrollment ≥ 60 % of aligned beneficiaries).	Directly links technology adoption to upside revenue and signals CMS' intent to "tie technology adoption to shared-savings programs."
Digital Quality Measure (dQM) Lift Payments	Provide a fixed per-member-per-year (PMPY) payment for the first performance year in which an ACO transmits digital quality measure (dQM) or electronic clinical	Offsets first-year integration costs for implementing FHIR bulk

	quality measure (eCQM) files that include patient-generated health data (PGHD).	export and device feeds into analytics sandboxes.
Prospective Care-Management Credits	Allow ACOs that deploy certified care-navigation apps to bill monthly CCM/TCCM-like codes without patient cost-share.	Rewards longitudinal digital touchpoints that reduce ED visits and readmissions.
Bonus Points in Benchmark/Re-contracting Scoring	Award points for documented reductions in chart-review hours through AI/natural language processing (NLP) abstraction or automated prior authorization.	Aligns provider-burden relief with MSSP re-enrollment decisions.
Rural Impact Multiplier	Apply a weighting factor (e.g., 1.1X) to shared-savings calculations for digitally engaged patients in rural/frontier ZIP codes or in the lowest Area Deprivation Index (ADI) quintile.	Addresses broadband and literacy barriers identified in frontier counties.

Obstacles Limiting Broader Adoption

1. **Data Interoperability Gaps** – Many ACOs lack software and processes to ingest device and app data into their data warehouses, despite availability of FHIR-bulk-export and TEFCA frameworks.
2. **Provider Workflow Burden** – Digital tools often create additional work required for completion. Initial case studies suggest that NLP abstraction can reclaim chart-review time, but such capabilities are not yet universal.
3. **Connectivity & Digital Equity Challenges** – Rural beneficiaries face broadband limits, and onboarding support for devices and multilingual apps remains uneven.
4. **Uncertain ROI / Payment Codes** – Remote-monitoring CPT codes do not fully cover newer internet-of-things (IoT) and prescription digital and AI-driven therapeutics, making the business case unclear for ACOs.
5. **Evidence Standards for PGHD** – It is unclear whether patient generated health data can be counted toward quality benchmarks without clear CMS guidance on data and measure validity.
6. **Stark Law and Anti-Kickback Statute Enforcement** – Originally designed to combat fraud, these laws hinder care coordination (e.g., value-based referrals, digital health integration).

VB-2. How can key themes and technologies such as artificial intelligence, population health analytics, risk stratification, care coordination, usability, quality measurement, and patient engagement be better integrated into APM requirements?

Artificial Intelligence Integration:

- Establish AI-enabled solutions for quality measure automation that reduces provider documentation burden while improving accuracy of risk adjustment and quality reporting

- Develop AI-powered care gap identification solutions that automatically flags missed preventive services, medication adherence issues, and care coordination opportunities
- Create AI-assisted clinical decision support integrated directly into APM quality improvement workflows
- Integrate AI-enable patient education into lifestyle modification programming (e.g., AI programs to match specific lifestyle change recommendations to identified chronic diseases).
- Implement predictive analytics for high-risk beneficiary identification to enable proactive interventions before costly events occur

Population Health Analytics:

- Use QIN-QIOs to publish population health dashboards across all APMs showing real-time quality metrics, cost trends, and beneficiary outcomes
- Require risk-stratified reporting that demonstrates how APMs are addressing different patient populations and social risk factors
- Establish benchmarking capabilities that allow APMs to compare their population health outcomes against similar organizations
- Integration with social determinants data to identify community-level intervention opportunities

Enhanced Care Coordination Requirements:

- Real-time care team communication platforms that automatically notify all providers when high-risk patients have emergency department visits or hospitalizations
- Standardized care plan sharing across all providers within an APM using FHIR-based APIs
- Care transition protocols with mandatory technology-enabled handoffs between care settings
- Patient engagement platforms that keep beneficiaries actively involved in their care coordination

Quality Measurement Integration:

- Automated quality measure calculation using bulk FHIR data exports to eliminate manual provider reporting
- Real-time quality dashboards showing APM performance against benchmarks with drill-down capabilities to individual provider and patient levels
- Outcome-based quality measures that leverage AI to identify true clinical improvements rather than just process compliance

Usability and Provider Adoption

- Simplified technology requirements that focus on API-enabled interoperability rather than specific software functionality
- Flexible implementation pathways allowing smaller or rural APMs to meet requirements through shared technology platforms. Technical assistance for small/rural APMs can be provided by QIN-QIOs.
- Technical assistance programs providing ongoing support for technology implementation and optimization
- User experience standards ensuring all APM-required technology meets accessibility and usability benchmarks

Patient Engagement Technology

- Mandatory patient portal integration with APM quality improvement initiatives
- Digital health product recommendations based on individual patient risk profiles and care gaps
- Patient-generated health data integration from wearables and remote monitoring devices
- Culturally appropriate engagement tools addressing social and non-health-related drivers of disparate outcomes.

The QIN-QIOs' unique value would be emphasizing practical implementation strategies that reduce provider burden while improving patient outcomes, leveraging their experience with quality improvement and provider support leveraging shared technology platforms, supporting phased rollouts approach, building out outcome-based incentives rewarding successful APMs and providing technical assistance and training programs.

VB-3. What are essential health IT capabilities for value-based care arrangements?

a. **Examples (not comprehensive) may include: care planning, patient event notification, data extraction/normalization, quality performance measurement, access to claims data, attribution and patient ID matching, remote device interoperability, or other patient empowerment tools.**

Capability	Why It Is "Essential" in VBC Context
IT Project Management (multi-disciplinary goals, role assignments, documented implementation plans)	Enables proactive data, measure and software integration.
Real-Time Patient Event Notifications	Reduce avoidable admissions and expedite post-discharge follow-up, a core MSSP metric.
Data Extraction, Normalization & Analytics Sandbox (with FHIR bulk-export ingestion)	Produces computable datasets for digital quality measurers (dQMs) and risk-stratification
Digital Quality-Performance Measurement (eCQM / dQM automation)	Automates reporting, reduces manual abstraction hours, and aligns with CMS move toward all-digital measures by 2027.

Access to Near-Real-Time Claims & Encounter Data	Gives ACOs visibility into out-of-network spend and ED use; critical for downside-risk models.
Attribution & Patient-ID Matching Services (MPI, probabilistic/deterministic)	Ensures beneficiaries are correctly aligned for both financial reconciliation and quality scoring.
Remote Device Interoperability & Command APIs	Supports RPM, SAMD, and prescription digital therapeutics integration into EHRs and care-navigation apps.
AI-Enabled Burden Reduction (prior-auth, NLP chart abstraction)	Demonstrated ability to "quantify chart-review hours saved," freeing clinical staff for higher-value tasks.
Patient Empowerment & Equity Toolset (low-literacy UI, multilingual, broadband-adaptive)	Addresses digital-equity gap highlighted for rural/frontier areas.

b. **What other health IT capabilities have proven valuable to succeeding in value-based care arrangements?**

VB-4. What are the essential data types needed for successful participation in value-based care arrangements?

- **Core Clinical EHR Data** – Problems/diagnoses (SNOMED), labs (LOINC), medications (RxNorm), vitals, radiology, and structured care-plan elements.
- **Administrative & Claims Data** – Medicaid & Medicare Advantage encounter data; Part A/B institutional & professional claims; Part D pharmacy.
- **Patient-Generated Health Data (PGHD)** – Remote patient-monitoring device streams (BP, glucose, heart rate, pulse oximetry, weight, sleep/activity), passive IoT sensors, app-captured patient-reported outcomes (PROs).
- **Social Drivers of Health (SDOH)** – Individual and community-level data such as Area Deprivation Index (ADI), provider shortage areas, food/housing insecurity screens, transportation access.
- **Behavioral & Mental-Health Assessments** – PHQ-9, GAD-7, substance-use screening, therapy adherence.
- **Longitudinal Care-Plan Status & Task Completion** – Indicates gap closure and goal attainment across providers.
- **Quality-Measure Artifacts** – eCQM numerator/denominator data, risk-adjustment variables, attribution files.
- **Provider Network & Cost Reference Files** – Fee schedules, contract terms, episode definitions needed for real-time financial forecasting.
- **Imaging & Diagnostic Reports (structured) and Unstructured Clinical Notes** – For advanced analytics (AI-augmented abstraction) and comprehensive risk stratification.